

**KENYATTA UNIVERSITY
DEPARTMENT OF PHILOSOPHY AND RELIGIOUS STUDIES
UCU 111 CRITICAL AND PROBLEM SOLVING**

INTRODUCTION: PHILOSOPHY

Meaning

Etymological (Literal Meaning)

- Philosophy derived from the Greek words; *philo*– love and *sophia*-wisdom
- Implies that philosophy is a thoughtful and creative response to the exigencies of life
- It is the search for answers, for understanding, for knowledge, it begins in wonder, curiosity, doubt
- Is an attempt to ask and answer fundamental questions of human life and the world in which we find ourselves.

General Sense of Philosophy (Popular Sense)

- A person's attitude towards certain actions, activities e.g. a person's philosophy of life
- A people's unique way of looking at the world; a people's world view. This would be sum total of their fundamental beliefs, values, practices, norms and so on.
- A general outlook on what constitutes an ideal society e.g. national philosophies such as Harambeeism, Nyayoism, Ujaamaism, African humanism, consciencism.
- Thus all people and cultures can claim to have a philosophy to the extent that they have some ideas about; physical objects/world; man; meaning of life and death; good and bad; right/wrong; beauty and ugly etc. These idea or views may be a result of deliberate thinking or may be largely a result of convention.

Technical Philosophy/Branches of philosophy

- It is a specialized area of inquiry distinguished by its experts, language, method, content
- Traditionally, Western philosophy is divided into four main branches: epistemology, metaphysics, logic and axiology.

-**Epistemology** is the theory of knowledge; it is an inquiry into the nature, validity, criteria, and sources of knowledge.

-**Metaphysics** deals with speculative questions in such areas as; cosmology (study of the universe), ontology (study of being), theodicy (inquiry into supernatural claims), philosophy of mind or rational psychology (inquiry into the natures of the mind and its relations to the body).

-**Logic**-the study of reasoning-focuses on rules, principles of reasoning by which we distinguish good from bad reasoning.

-**Axiology** is the philosophical study of value in general, for example artistic, social political and ethical values. Ethics, Aesthetics, social political philosophy are sub-branches of Axiology.

Methods of Philosophy

CRITICAL method

- Accepts only what is well grounded
- Bases convictions, beliefs, and actions on well-reasoned grounds
- Frees us from slogans, hypocrisy Philosophy is

REFLECTIVE

- Scrutinizes common sense view to discover the underlying basis and validity
- Inquiry -360% perceptibility of issues, not straight – jacket
- Open minded view of worlds
- Questioning

RATIONAL

- Relies on power of reason to see the how we can stretch human reason to answer fundamental questions
- Relies On Criteria, rationale, or Standards

ANALYSIS

- Breaking thinking into its constituent part,
- To understand the parts that constitute the issue at hand

SYNTHESIS

- Involves construction as explanation by combining different idea, perspectives to come up with a new and richer understanding.

DIALECTICAL METHOD

- It is a higher level thinking, involving the progression of two opposing ideas or perspective to a synthesis
- It emphasizes logical progression of thought or argumentation

SPECULATIVE METHOD

- Trying to find a coherent explanation for reality and experience for example of metaphysical problems such as the genesis of the universe, human existence of God and so on

Examples of Definitions of Philosophy

-Philosophy is a method of reflective thinking and reasoned inquiry; as a method it is careful and accurate thinking, philosophical method is reflective and critical

-Philosophy is an attempt to gain a whole view of the world; philosophy takes a holistic view of reality

-Philosophy is a personal attitude towards life and universe; a philosophical attitude is one that is searching, critical, open-minded, tolerant

- Philosophy is a group of problems as well as theories about solutions to these problems
- Philosophy rationally examines the basic problems that affect man and world he lives in.

Relevance of Philosophy

- 1 Develops in individuals logical, coherent, systematic, consistent, clear thinking, critical thinking
- 2 Such thinking enhances our rational capacity which in turn enhances decision making, problem solving.
- 3 It also liberates the mind from narrow minded to open minded by enlarging one's scope of perception and awareness.
- 4 Philosophy helps to come to terms with meaning and significance of human life in its existential conditions.
- 5 It helps to ground values and convictions on rational grounds or foundations, thereby sharpening our moral sensibilities.
- 6 Philosophy raises and attempts to provide answers and explanations to some of the most complex, troubling questions of human existence e.g. death, destiny of man etc.

THINKING

Broadly

- Thinking is characterized as a mental process or activity, an activity of the mind
- Can range from simple to complex operations of the mind e.g. daydreaming, remembering, doubting, questioning etc.

Thinking serves several purposes, for example:

- Cognitive -to gain knowledge and understanding
- Contemplative/evaluative- to ascertain whether actions are right/wrong e.g. in moral reasoning
- Deliberative -to solve problems, make decisions, resolve situations.

REASONING

- Sometimes used interchangeably with thinking.
- Reasoning is the intellectual dimension of thinking.
- Reasoning occurs when the mind draws or infers conclusions on the basis of reasons or premises; we draw conclusions whenever we make sense of things.
- We draw conclusions about many things all the time, hence to reason well – we must scrutinize the process we are using.

LEVELS OF THINKING

The following are three broad levels of thinking

Pre- conventional thinking

Individuals at this level are characterized by the following;

- Operate on Animal- like instinct
- No rationality
- Not in control of thinking
- No sense of right and wrong, cannot sense of moral value
- Operate on reward/ punishment principle
- Hence individuals are not responsible for their actions.

Conventional thinking

Individuals at this level are characterized by the following;

- Emergence of rationality
- They engage in rational thought
- They have formed the concept of right and wrong

- They can provide reasons to account for their thinking
- However, the individual is limited to thinking within confines of the socially, culturally popularly approved and accepted ways of thinking
- Thus social approval provides the rationale for action and conduct and thinking
- One defers to society, one sees oneself in terms of society, hardly questions society
- Conformity to authority is thus the hallmark of thinking at this stage.

The problem with this is that, society controls the individual's thinking, easily takes advantage of individuals and may:

- Lead them astray
- Lead to the irrationality of mass thinking
- Society tends to be paternalistic and authoritarian

Post- Conventional (Autonomous) thinking

This level is also called the autonomous stage and is characterized by;

- Autonomy of thought and action.
- Originality of thought
- Grounded in the belief that we ought to think for ourselves
- Characterized by reflective thinking
- Individual is self- directed as opposed to being other directed
- Individuality fully developed
- Individuals fully reflect on socially, culturally, popularly accepted values, norms often radically questioning their underlying basis.
- Individual manifest a highest degree of creativity, innovation and criticality
- Such individuals transcend society, hence they can correct and change society.

CRITICAL THINKING

Wolff defines critical thinking as habit of carefully reasoned inspection of the way we evaluate, judge and act with the aim of making ourselves wiser, more reflective and therefore better men and women. Thus critical thinking is;

- A higher level thinking, as a mental activity it is self- directed activity.
- It is guided by reason; it seeks evidence for claims.
- It is reflective, it attempts to go deeper into things and avoid deceptive situations.
- It is thinking that facilitates good judgment, one is able to make more rational judgment
- It is thinking that is purposive, intentional, deliberate and focused not merely accidental and spontaneous

A critical thinker is thus one who among other characteristics is;

- An autonomous thinker (thinks for oneself)
- Does not accept what has not been proved and well demonstrated
- Doubts, questions, takes nothing for granted and acts on conviction
- Strives to distinguish between appearance and reality

CRITICALITY AND CREATIVITY

Criticality and creativity are two sides of the same coin.

They are inextricably connected, intertwined and reinforce each other to improve reasoning

Creativity

- Generative thinking
- Produces, ideas materials
- Proposes – alternatives,
- Options, solutions, decisions

If critical thinking judges something as being wanting, it means, creativity must generate further information.

Criticality

- Evaluative
 - It assesses
 - Critiques what creativity generates
- Points out strengths and weaknesses
- Defends by giving justification

CREATIVE THINKING

Reasoning that is characterized by imagination, innovation and originality in the generation of ideas.

Imagination – ability to be creative, new interesting ideas or insights

Innovative – ability to be inventive, introduce new ways of thinking

Productive – generation of something new and of value to human wellbeing

Improvement – ability to better human wellbeing

Creativity implies – reasonableness, independence, handwork – which enable individuals to solve problems and make decisions and choices

WHY STUDY CCT

- Equips us with reasoning skills which improve our reasoning capabilities such as clarification, illustration, analogies.

- Develops intellectual standards such as accuracy, precision, depth, relevance which enable us to assess the quality and value of thinking
- Cultivates intellectual virtues/dispositions such as autonomy, open-mindedness which help to transform our thinking.
- Facilitates responsible decisions making and judgments, to improve our capacity to solve problems, make decisions.
- Enhances communication skills
- Enhances logical, systematic, coherent reasoning
- Enhances our power to persuade and convince

OBSTACLES/BARRIERS OR IMPEDEMENTS TO CRITICAL THINKING

There are many thought impediments that may prevent a person from thinking clearly. Francis Bacon calls them idols of the mind. The following examples prejudice, habits, propaganda, authoritarianism, paternalism, indoctrination, sociocentrism

PREJUDICE

-A prejudice is a pre-judgment or a mental bias

-A prejudice usually rests on emotional grounds and tends to be in line with a person's self-interest, pride or comfort

-A prejudice impedes thinking by making one;

- To rationalize i.e. to find reasons to continue to believe what we value
- To recognize only evidence that is favorable e.g. to our gender, tribe group etc.
- To ignore or minimize the part of evidence that is unfavorable to us
- To overestimate parts of the evidence that is favorable to us
- To see things from our limited point of view
- To be unable to draw accurate conclusions

HABIT

-A habit thinking or doing things repetitively often without careful thought, reflection and criticism

- Thus through habits we learn to do things only in one particular way

-Habits hinder our thinking by:

- Impeding one's capacity for reflection and criticality
- Making us dogmatic
- Making us blind to alternatives

- Making us recognize only evidence that is familiar to us hence making it difficult to deal with new and unfamiliar situations
- Making us to have a superficial view of things

PROPAGANDA

-Propaganda is the deliberate tainting, manipulation or distortion of information in order to achieve a certain predetermined end or goal e.g. political, commercial, military, and religious, gender.

-The propagandist first tries to arouse in the people some strong emotion or desire and then through suggestion present a line of action that appears to satisfy that emotion or desire

-Propaganda inhibits thinking by:

- Controlling and manipulating our thinking
- Making it difficult for us to think for ourselves
- Leading us into biased and inaccurate conclusions
- Encouraging mass thinking-mass media used –popular device in advertisers

AUTHORITARIANISM

-This the uncritical and or blind appeal to authority, leading to uncritical and unreflective acceptance of what emanates from authority

-This is normally grounded in the false believe that knowledge is validated or guaranteed by authority

-Usually we are led astray by the prestige of authority and even fail to realize when they speak outside their area of competence

-Many people rely on authority because either they have little confidence in their own thinking or they are intellectually lazy

-Authoritarianism impedes thinking by:

- Making us accept things uncritically and unreflectively
- Making us to accept things without question
- Blocking progress in further thinking and investigation
- Making us surrender ourselves to others to think for us.

REASONING SKILLS AND TOOLS

THINKING MAP

- This is a list of key questions that one should ask when weighing up reasoning or an argument.
- These key questions can be divided into three sets, analysis, evaluation and transformation.

TOOLS OF ANALYSIS

They are also called elements of reasoning-they cut across all disciplines
They help us to separate thinking into its component parts
They enable us to recognize flaws or errors in reasoning.
These tools therefore guide one in understanding what is being said or what is being argued

They are captured by the following questions among others:

1. What is the main conclusion?
2. What are the reasons evidence or premises?
3. What is assumed and what is the context?
4. Are there terms or concepts that require clarification?

There are eight tools of analysis as follows:

Purpose

- Reasoning is not random, but selective, based on goals, desires, needs and values
- Our reasoning is an integral part of our patterned way of acting in the world
- To understand reasoning we must understand the function that it serves.
- Most of what we are after is not obvious to us
- Hence critical thinking helps us to raise our goals to level of conscious realization

Point of View

- Reasoning has focus or orientations
- Reasoning is focused on something from an angle both can change
- To understand thinking we must understand the perspective. For example; politically, scientifically, poetically, philosophically, conservatively, liberally, religiously, secularly and so on or combination of different perspectives.

Concepts/Theories

- These are general categories or ideas by which we interpret, classify and group information used in reasoning
- Each discipline has its own set of concepts, technical vocabulary and theories that facilitate its thinking.

Question/Issue or Problem

- In thinking about ourselves and the world, we usually face questions we need to answer, problems we need to solve and issues we need to resolve
- To improve thinking we must learn how to pose the questions, problems and issues as these also affect the focus.
- For example: What can be done about the education system in Kenya? And what can educators do to ensure that students learn skills and abilities that help them function successfully on the job and in their daily decision making?

Data, Facts, Information

- The stuff about which we are basing our reasoning, that which supports our conclusion,
- To improve our reasoning we must assess the facts, information and data

Conclusion/Inference

- This is taking something we know (which we believe we know) and figuring out something else on the basis of it- to infer.
- For example if you walk past me without saying hello, I might infer or come to the conclusion that you are angry with me
- In life we continually make inferences about people, things, places events of our lives etc
- It is to consider how we draw those inferences.

Assumptions

- This is the starting point of reasoning, what is accepted or taken for granted by the speaker writer usually not stated explicitly on the basis of which we can figure out something else.
- For example you say he is a democrat he will support freedom of the press-the assumption is that democrats support freedom of the press.
- Or if you reason that someone who invites you to her room after a party at night to continue with this interesting conversation is interested in you romantically or sexually you assume that the only reason for going to someone's room late at night is to pursue a romantic or sexual relationship
- Any defects in the assumptions or presuppositions from which reasoning begins is a possible source of problems in reasoning

Implications

- Whenever we reason, implications follow, it means that to which our thinking is leading to or consequences of our thinking.

-For example if you call yourself a feminist you imply that you are in favour of a political, social and economic equality of the sexes. OR if you make a promise you imply that you intend to keep it

TOOLS OF EVALUATION

They are also called standards of reasoning

These tools that help us to assess, judge, justify and critique thinking

They help us to point out the strengths and weaknesses of reasoning

They guide one in deciding whether one should be convinced by the argument or reasoning or not.

They are captured by the following questions among others:

1. Are the reasons acceptable; what is the credibility of the source, is it consistent, does the source have vested interest or bias, is it from expertise?
2. Does the reasoning support the conclusion: how much support has been given, are there any other arguments or relevant considerations which may strengthen or weaken the case?
3. What is your overall evaluation of all the issues raised by the foregoing questions?

The key tools of evaluation include the following:

Accuracy

-To be accurate is to represent something in accordance with the way it actually is.

-People often misrepresent or falsely describe things especially when they have vested interest e.g. Advertisers, politicians, propagandists and so on.

-Good thinkers must question whether what they hear is true and accurate

Clarity

- This is a gateway standard; it helps to determine whether the point at hand is accurate or relevant

-For example we ask; could you elaborate on the point? Could you put it in another way? Could you give an illustration or an example?

-If a statement is unclear we cannot say anything about it because we do not know yet what it is saying.

Precision

-A statement can be both clear and accurate but not precise

-To be precise is to give the details needed for someone to understand exactly what is meant.

-In some situations specifics are essential for good reasoning yet in others they may not.

-For example we ask; could you give more details? Could you be more specific?

Relevance

- Something is relevant when it is directly connected with or bears upon the issue at hand.
- For example when it is pertinent or applicable to a problem you are trying to solve
- A statement can be clear, accurate, precise but not relevant
- Thus we ask; how is the idea connected to the question? How does it bear on the issue? How is it related to this other idea? How does your question relate to the issue we are dealing with?
- Thinking that is relevant remains on track.

Depth

- This thinking that gets beneath the surface of an issue or problem, to identify the complexities inherent in it and deal with those complexities in an intellectually responsible way.
- A statement can be clear, accurate, precise, relevant but superficial-lacking in depth
- For example; what should be done about HIV and AIDS among the youth in Kenyan universities, just say chill
- Thus we ask; how does your answer address the complexities in the question? How are you taking into account the problem in question? How are you dealing with the most significant factors in the problem?

Breadth

- This is to consider an issue at hand from every relevant point of view.
- A line of reasoning can satisfy all the standards considered above but lack breadth
- When multiple points are pertinent to an issue yet we fail to give due consideration to those perspectives we think myopically or narrowly; we do not consider alternative or opposing point of view.
- Humans are frequently myopic for many reasons: limited education, sociocentrism, selfishness, self-deception, intellectual arrogance
- Points of view that significantly disagree with our own often threaten us; may require that we reconsider our views or positions.

TOOLS OF TRANSFORMATION

They are also called intellectual virtues, traits, dispositions

These are good intellectual habits we acquire when we reason well.

Thus critical thinking inculcates these intellectual virtues which transform the way we reason

They deal with the question that helps us determine the value of reasoning well. The fundamental question is

1. What intellectual traits has one acquired or cultivated through critical thinking?

They are traits of mind and character that we cultivate when we reason well.

They are necessary for right action and thinking; the traits of mind and character essential for fair-minded rationality; the traits that distinguish the narrow-minded, self-serving thinker from the open-minded, truth-seeking critical thinker.

-These intellectual traits are interdependent of each other but best cultivated simultaneously.

-They cannot be imposed from without; they must be cultivated by encouragement and practice.

They perfect the intellectual activity of man. They include among others:

Intellectual Autonomy

-It involves rational control of one's beliefs, values, and inferences

- The ideal of critical thinking is to learn to think for oneself, to gain command over one's thought processes.

- Intellectual autonomy does not entail willfulness, stubbornness, or rebellion.

- It entails a commitment to analyzing and evaluating beliefs on the basis of reason and evidence, to question when it is rational to question, to believe when it is rational to believe, and to conform when it is rational to conform.

Confidence or Faith in Reason.

-Confidence that in the long run one's own higher interests and those of humankind at large will best be served by giving the freest play to reason by encouraging people to come to their own conclusions through a process of developing their own rational faculties

- Faith that (with proper encouragement and cultivation) people can learn to think for themselves, form rational viewpoints, draw reasonable conclusions, think coherently and logically, persuade each other by reason, and become reasonable, despite the deep-seated obstacles in the native character of the human mind and in society.

-Confidence in reason is cultivated through experiences in which one reasons one's way to insight, solves problems through reason, uses reason to persuade, and is persuaded by reason.

-Confidence in reason is undermined when one is expected to perform tasks without understanding why, to repeat statements without having verified or justified them, to accept beliefs on the sole basis of authority or social pressure.

Intellectual Courage

- The willingness to face and fairly assess ideas, beliefs, or viewpoints to which we have not given a serious hearing, regardless of our strong negative reactions to them.
- This courage arises from the recognition that ideas considered dangerous or absurd are sometimes rationally justified (in whole or in part), and that conclusions or beliefs espoused by those around us or inculcated in us are sometimes false or misleading.
- To determine for ourselves which is which, we must not passively and uncritically "accept" what we have "learned."
- Intellectual courage helps us to come to see some truth in some ideas considered dangerous and absurd and some distortion or falsity in some ideas strongly held in our social group.
- It takes courage to be true to our own thinking because questioning cherished beliefs is difficult, and the penalties for non-conformity are often severe.

Intellectual Empathy

- Imaginatively putting oneself in the place of others to genuinely understand them.
- We must recognize our egocentric tendency to identify truth with our immediate perceptions or longstanding beliefs.
- Intellectual empathy correlates with the ability to accurately reconstruct the viewpoints and reasoning of others and to reason from premises, assumptions, and ideas other than our own.
- This trait also requires that we remember occasions when we were wrong, despite an intense conviction that we were right, and consider that we might be similarly deceived in a case at hand.

Intellectual Humility

- Awareness of the limits of one's knowledge, including sensitivity to circumstances in which one's egocentrism is likely to function self-deceptively
- Sensitivity to bias and prejudice in, and limitations of one's viewpoint.
- Intellectual humility is based on the recognition that no one should claim more than he or she actually knows.
- It does not imply spinelessness or submissiveness; rather it implies the lack of intellectual pretentiousness, boastfulness, or conceit, combined with insight into the strengths or weaknesses of the logical foundations of one's beliefs.
- Intellectual Integrity

- Recognition of the need to be true to one's own thinking, to be consistent in the intellectual standards one applies, to hold oneself to the same rigorous standards of evidence and proof to which one holds one's antagonists,
- To practice what one advocates for others, and to honestly admit discrepancies and inconsistencies in one's own thought and action.
- This trait develops best in a supportive atmosphere in which people feel secure and free enough to honestly acknowledge their inconsistencies, and can develop and share realistic ways of ameliorating them.
- It also requires honest acknowledgment of the difficulties of achieving greater consistency.

Intellectual Perseverance

- Willingness and consciousness of the need to pursue intellectual insights and truths despite difficulties, obstacles, and frustrations
- Firm adherence to rational principles despite irrational opposition of others; a sense of the need to struggle with confusion and unsettled questions over an extended period of time in order to achieve deeper understanding or insight.
- This trait is undermined when teachers and others continually provide the answers, do students' thinking for them or substitute easy tricks, algorithms, and short cuts for careful, independent thought.

Intellectual Sense of Justice

- Willingness and consciousness of the need to entertain all viewpoints sympathetically and to assess them with the same intellectual standards, without reference to one's own feelings or vested interests, or the feelings or vested interests of one's friends, community, or nation
- It implies adherence to intellectual standards without reference to one's own advantage or the advantage of one's group.

ARGUMENTS

BY BRIAN SKYRMS

DEDUCTIVE AND INDUCTIVE ARGUMENTS/REASONING

Reasoning

It is the ability of the mind to draw/infer a conclusion from a set of assumptions or premises or reasons.

Statement/Proposition

Is sentence that makes a factual claim, i.e.

- It has truth value
- It is either true or false.
- It can be asserted or denied
- It can be given as evidence to support a certain claim/conclusion.

Premises

These are reasons given in support of the conclusion. They constitute the evidence given in support of a point of view or a conclusion in an argument. Premises are usually identified by words called premise indicators, such as: since, because etc.

Conclusion

This point of contention in an argument, it is the point that we seek to establish in an argument. It is a claim that we seek to proof in an argument. In an argument the conclusion is usually identified by words called conclusion indicators such as: therefore, consequently etc.

Argument:

Brian Skyrms defines an argument as a list or group of statements one of which is called conclusion, the rest of which are called premises.

Logic

Brian Skyrms defines logic as the study of the evidential link between premises and the conclusion in an argument.

DEDUCTIVE REASONING

Traditionally, it is defined as thinking that proceeds from general principles or theories to specific a conclusion.

According to Skyrms the main distinguishing feature of deductive thinking does not however lie in the fact that it proceeds from general to particular but in the relationship of **necessity** between the premises and the conclusion. Hence deductive reasoning is characterized by the following;

- The truth of the premises guarantees the truth of the conclusion
- The conclusion is contained or implied by the premises
- The premises make the conclusion inevitable or unavoidable.
- If you accept the premises you must accept the conclusion
- If the premises are true then conclusion must be true, because the conclusion is usually part of the premises.
- Deductive reasoning measures the validity in arguments. A valid argument is one where it is impossible for the conclusion to be false while its premises are true. It is the strongest possible argument.

Thus Brian Skyrms defines deductive reasoning as follows;

-An argument is deductively valid if it is **impossible** that the conclusion is false while premises are true.

-These arguments are the strongest arguments ever and are described as **valid**. Deductive logic is concerned with tests of deductive validity.

-This kind of reasoning applied is mainly in pure logic and pure math.

EXAMPLES

P1. All Professors are absent-minded

P2. Dr. Wise is a Professor

Therefore: Dr. Wise I absent-minded

P1. All human beings are mortal

P2. Socrates is a human being

Therefore: Socrates is mortal

P1. If you are a police officer, you cannot be a thief

P2. John is a police officer

Therefore: John cannot be a thief

INDUCTIVE REASONING

Traditionally, said to be reasoning that proceeds from particular, specific cases to general conclusion.

All arguments cannot meet the rigorous standard of inference seen in deductively valid arguments.

Thus when an argument is not deductively valid but the premises provide good evidence for the conclusion the argument is said to be inductively strong or cogent

The strength or cogency of an argument depends on how much evidential support the premises give for the conclusion.

Therefore, in these arguments the link between premises and the conclusion is based on **probability**. That is;

- The premises give support to the conclusion but do not guarantee the truth of the conclusion
- The premises make the conclusion probable, likely
- They are therefore evaluated on the basis of inductive strengths. They measure cogency in arguments. A cogent argument is an argument where the premises reasonable support the conclusion.
- Therefore, inductive arguments vary in strength, depending on how probable the conclusion is given the premises; the higher the probability, the more cogent the argument is.
- The conclusion asserts more than what is contained in the premises.
- The conclusion ventures beyond the claims made in the premises since we can envisage a situation where the premises could be true but the conclusion is false.
- Hence inductive arguments take the risk of proceeding from true premises to a false conclusion. This risk is referred to as inductive risk.

Hence defined by Brian Skyrms as;

- An argument is inductively strong if it is improbable that the conclusion is false while its premises are true.

-Inductive logic is thus concerned with tests of measuring inductive probability-the degree of inductive strength depends on how improbable it is that the conclusion is false while the premises are true.

-This reasoning is applied mainly in science and can be used to;

Forecast
Discoveries
Generalization
Predictions.

EXAMPLES

P1. George is a man
P2. George is a hundred years old
P3. George has arthritis

Therefore: George will not run a four-minute mile tomorrow

P1. The Metrological department have said it will rain today
P2. In the streets, people are carrying umbrellas
P3. In the sky there is heavy dark clouds overcast

Therefore: It will rain

NOTE:

TESTS FOR INFERENCE

-There must be some reasonable, secure connection between reasons and the conclusion if the premises are to justify the conclusion. A link we can understand and accept.

-Even if the premises are true or acceptable but there are other relevant considerations that can make the conclusion to be false, or unacceptable, then the inference is not good.

FALLACIES (BY IRVING COPI)

-A fallacy is a mistake or error in reasoning; Fallacies are simply mistakes that occur in arguments that affect their cogency and validity.

-Arguments that may seem to be correct but upon careful examination they are not. -Fallacious arguments are normally deceptive. In fact, the Latin word for fallacy is *felereor* or *falacia* which means to deceive.

-In other words, fallacies are bad arguments that masquerade or pretend to be good ones; they are counterfeits of good arguments.

- Fallacies are not mere conceptual problems they have far-reaching implications as they may:

- Misrepresent
- Prejudice leading to animosity
- Used for sloganeering and propaganda to cloud our minds and divert us from the real issues.

-Fallacies deceive because they appear good by appealing to our emotions, prejudice, self-interest or blind faith.

Note: A fallacy is a logical error (in reasoning) as it involves the violation of the conditions of rationally acceptable inference or drawing of a conclusion. Hence whenever we reason invalidly or irrelevantly, accept premises we should not or fail to make appropriate use of relevant facts at our disposal, we commit a fallacy.

Classification of Fallacies

Broadly speaking, fallacies can be divided into two major categories – **Formal and Informal fallacies**.

Formal fallacies are also known as logical fallacies.

- These mistakes arise through the violation of the logical rules, which govern correct deductive arguments.
- They occur when we misapply a valid rule of inference thereby contravening the formal structure of a valid argument.
- They are invalid deductive arguments resulting from bad deductive reasoning.
- They can, therefore, be identified by a mere inspection of the argument form.

Informal fallacies

- Unlike the formal fallacies, the informal mistakes do not relate to the form of the argument in which they occur.
- They are identified through an analysis of the content of an argument.

These fallacies are subdivided into **four** broad categories:

1. Fallacies of relevance

They occur when the premises of an argument have no bearing upon its conclusion. That is they occur in arguments whose premises have no logical relevance to their conclusion. In addition, such fallacies often involve a distractive element which diverts attention away from this very problem. (Such arguments are often referred to in Latin as **non sequiturs** which means, it does not follow).

The irrelevance here is logical and not psychological. They are deceptive because of the psychological relevance which is easily or often confused with logical relevance. This then defeats the purpose of logic namely, the observance of relevance between the premises and their conclusion in any given argument.

- a) **Ad hominem** (attacking the person) arguments try to discredit a conclusion (claim, proposal etc.) by attacking its proponents instead of providing a reasoned examination of the conclusion (proposal) itself. They are divided into several variants with the most pervasive being *ad hominem* abusive and circumstantial.

Ad hominem abusive arguments attack a person's age, character, family, gender, ethnicity, appearance, dress, personality, economic status, behavior or professional,

political, or social affiliations. They employ the principle of transference i.e. what is true in psychology is also true in logic. They are also called the genetic fallacies.

They also evoke an attitude of disapproval towards a person. The implication is that there is no reason to take the person's view seriously.

Circumstantial – use the unique or special circumstances of a person to discredit his reasoning. e.g., guilt by association hence sometimes referred to as “poisoning the well” argument,

b) Ad populum arguments (appeal to the people, masses, multitude or gallery) occur when we infer a conclusion merely on the grounds that most people accept it. It tries to invoke the band wagon effect, which asks us to join forces with others often irrationally. Hence assumes that because something is popular, it is therefore good, correct or desirable. This fallacy is evident in populist political speeches demagoguery. Much of advertising is also based on this fallacy as well. ‘The car in front is always a TOYOTA,’ ‘500,000 Kenyans cannot go wrong – AAR’. They usually associate things with people that arouse strong emotional approval from the masses. ***Note: popular opinion cannot be taken as a proof that an idea is right or wrong. Remember everyone believed that the world was flat and that the Earth was at the center of the universe!***

c) Red Herring is a fallacy that introduces an irrelevant issue to divert attention from the subject under discussion. A red herring is an extraneous matter used purely to divert attention away from the issue posed by an argument. Because it is irrelevant, it contributes nothing to an argument, though it misleads its audience into thinking otherwise. Red herring enable those who use them to mask other defects in their arguments and thus to evade the real issue.

-The name of this fallacy comes from an old trick used by farmers in Europe to keep hunters and their dogs from galloping through crops. By dragging a smoked herring with a strong odor along the edge of their fields, farmers threw the dogs off the track by destroying the scent of the fox.

d) Ad Vericundiam (appeal to authority) occurs when we accept (or reject) a claim merely because of the prestige, status, or respect we accord its proponents (or opponents) e.g. Testimonials – exemplified by celebrities who appear on adverts and commercials endorsing products, services or brands of goods. Mariga – UAP, Patrick Njiru with Panadol etc.

e) Ad Misericordiam (appeal to pity). This occurs when we arouse in the audience sense or feeling of pity or sympathy and appeal to it to win argument. This is used mainly by defense lawyers even the prosecution in cases. The legal systems encourage this.

2 Fallacies of Presumption

They arise in arguments where the premises presume what they purport to prove.

a) **False dichotomy:** Also called the *Either or* fallacy

This fallacy forces one to choose between two alternatives when more than two alternatives exist. It oversimplifies a complex issue by reducing it to a simple either or choice. The alternatives it presents do not exhaust all the possibilities

Either you are with us or you are against us

You are not with us. Therefore, you must be against us.

b) **Complex question:** Also called the fallacy of many questions, fallacy of interrogation

It consists of asking questions in such a way that any single answer involves or implies other answers as well. In a complex question usually two or more questions are rolled into one. In the case of two questions the second presupposes that a definite answer has already been provided to the unasked question. E.g. *did your sales increase as a result of your misleading advertisement? Yes or No*

These devices are exploited by lawyers in cross-examination to have the accused confused, incriminate themselves. Also popular with propagandists

c) **Slippery slope fallacy** occurs when the conclusion of an argument rests upon an alleged chain reaction, suggesting that a single step in the wrong direction will result in a disastrous or otherwise undesirable outcome. In other words, it is a fallacy which assumes that taking a first step will lead to subsequent steps that cannot be prevented. *It takes its name from the image of a boulder rolling uncontrollably down a steep hill. Once the boulder gets started, it cannot be stopped until it reaches the bottom.* A person who commits this fallacy assumes that taking a first step will lead inevitably to a second step and so on down the slope to disaster; without providing evidence to support such a claim. However, to assume that all the later steps will occur without proving that they will is to commit the slippery slope fallacy. E.g. *Passing laws to control the amount of violence on television is the first step in a process that will result in absolute government control of the media and a total censorship of all forms of expression*

3 Fallacies of Ambiguity

Occur in arguments whose formulations contain ambiguous words or phrases and whose meaning(s) shift and change in the course of the argument and thus render them fallacious. They are subtle; principally involve the use of language, hence also called fallacies of clearness

a) **Equivocation**

Equivocation arise when we confuse the different senses a word or phrase may have using them in different senses in the same context. In the context of an argument this is called fallacy of equivocation. The main aim is either to hide the truth or to mislead the listener. Examples:

The end of a thing is its perfection. Death is the end of life. Hence death is the perfection of life.

It is silly to fight over mere words. Discrimination is just a word. Therefore it is silly to fight over discrimination.

Men are the only rational creatures on planet earth. No women are men. Therefore, no women are rational

- b) **Amphiboly.** In language this occurs when we cannot readily determine the meaning because of either the words being arranged awkwardly or loosely. In the context of argument fallacy of amphiboly is committed when it is stated as premise on interpretation which makes it true and conclusion is deduced on the basis of interpretation that makes it false. Example:

Croesus king of wanted to go to war with king Cyrus of Persia. He went to consult the oracles who told him; **if you go to war with Persia you will destroy a mighty kingdom**

- c) **Accent** refers to emphases that generate multiple (and often misleading) interpretations. e.g. Newspaper headlines, contracts and deceptive entry forms are frequent sources of fallacies of accent.

d) **Fallacy of Composition**

First it occurs when arguing from the properties of the parts of a whole to the parts of the properties of the whole itself e.g. every parts of a machine is light therefore the whole machine is light

Second it occurs when arguing from the properties held by individual elements of a collection to properties held by the collection or the whole itself e.g. that a bus uses more fuel than a a saloon car, therefore buses use more fuel than saloon cars

e) **Fallacy of division**

First, it is committed when arguing that what is true of the whole must be true of its parts eg. Kenyatta university is a very important institution, and because must Kamau works in Kenyatta university he must be very important

The second variety is committed when one argues from the properties of the collection or a whole to the properties of the elements themselves. For every example Kenyatta University offers courses in Engineering, Medicine, Law,

therefore, each or any student of Kenyatta university studies medicine, law, Engineering.

4 Inductive Fallacies

Occur when the inductive probability of an argument (i.e. the probability of its conclusion given the premises) is low or at least lower than the arguer thinks it is.

- a) **Hasty generalization** means inferring a conclusion about an entire class of things from inadequate knowledge or unrepresentative cases of some of its members. Hasty generalizations are usually fallacious statistical/inductive generalizations and stem from biased, unrepresentative or inadequate sampling techniques. Pollsters, Surveys, marketing gimmicks. Example;

My Pastor is insincere therefore, Christians are insincere.

b) **Faulty analogy**

Analogical reasoning is a powerful tool in reasoning. However, analogical reasoning depends quite sensitively on the degree and relevance of the similarity.

Why Are Fallacies Deceptive?

- 1) Carelessness and being inattentive to detail
- 2) Use of the heart or emotions to think. They appeal to our emotions, prejudice, self-interests or blind faith.
- 3) Ambiguity in language – semantics/formulation of the argument
- 4) Not acknowledging the limits of our knowledge, extent of our ignorance
- 5) Mental fatigue/tiredness
- 6) Generally, by not being critical and creative

NOTE:

To help us avoid being victims of fallacious reasoning we need to be guided by the thinking map used in evaluating arguments and reasoning

DECISION MAKING AND PROBLEM SOLVING

Decision making and problem solving are inextricably connected or intertwined:

- A problem is an obstacle to overcome
- A decision is a solution to the obstacle

Therefore, every domain of decision making is also a domain of problem solving

Every decision impacts our problems; it can either minimize or enhance

Need for Rational Decision Making and Problem Solving

When the pattern of decision making is rational:

- We live a rational life.
- Maximize our chances of happiness
- Enhance chances of successful living and fulfillment.

Role of Critical Thinking in Decision Making and Problem

- Enhances the rationality of our decisions and our solutions to our problems.
- Raises decision-making and problem-solving to the level of conscious and deliberate choice.
- Enhances evaluative and analytic skills
- Enhances creative skills
- Enhances reasoning skills
- Helps us to avoid short-term and spontaneous decision- making and problem- solving

DECISION MAKING

To live is to act. To act is to decide. Everyday life is an endless sequence of decision making.

Some of the decisions are small and inconsequential and some are large and life determining.

Categories of Decisions we make

- 1) Basic Human needs-we all make choices and decisions as to how to satisfy these needs
- 2) Chosen Values-we also make choices and decisions in relation to the values that we impose upon ourselves.
- 3) Implications to the wellbeing of others.

Note:

- Decisions that undermine and harm the wellbeing of others are unethical
- Decisions and values that undermine and harm ourselves are irrational and unethical

Common Patterns of Irrational/Unethical Decision Making

1. Deciding to behave in ways that undermine our welfare.
2. Not to engage in activities that contributes to our long term welfare.
3. Deciding to behave in ways that undermine another's welfare.
4. Deciding to associate with people who encourage us to act against our own welfare and others welfare.

Main Reason for Such Self-Defeating/Self-Harming Decision:

- Immediate gratification and short-term gain i.e. we make decisions with immediate gain and the short run uppermost in our minds
- Indeed, our mind seems to be ‘wired’ for immediate and short- run gratification.
- We tend to be driven by hedonistic tendencies, seeking pleasure

BIG DECISIONS

There are two kinds of decisions to learn to watch for in one’s life.

1. Those that have obvious long-term consequences e.g. basic parental decisions, career choices, choices of mate etc.
2. Those whose long term consequences must be discovered e.g. Implications of our daily habits, behaviors, values

Four Ways to Sound Decision – Making

- 1 To recognize that you face in important decision.
 - This is to enable one avoid making the decision subconsciously, to let the un-thought decisions creep into our lives
- 2 To accurately identify the alternatives. Failure arises in two ways:
 - When we think something is an alternative when it is not (thinking unrealistically)
 - When we fail to see an alternative (thinking narrowly)
- 3 To logically evaluate the alternatives
 - This needs to be done in line with our goals, purposes
- 4 To have the self-discipline to act on the best alternatives
 - This emphasizes the need to remain rational even in the face of temptations to make emotional decisions etc.

Each of these factors presents potential problems to the thinker.

Skilled Decision-Making

- Put more time in decision making as it costlier to deal with the negative effects of a bad decision.
- Being systematic and focused on the decision.
- Dealing with one major decision at a time.
- Developing knowledge of one’s ignorance(open-mindedness)

Thinking Map in Handling Decisions and Recommendations

1. What makes this decision necessary? Objectives
2. What is recommended and on what grounds?

3. What are the options/ alternatives?
4. What are the possible implications and consequences of the various options – and how likely are they?
5. How important are these consequences for all those affected?
6. When I compare the alternatives in the light of their consequences, which is the best? Or the recommended course best?
7. How can I carry through my decision? Contingency plans?

PROBLEM-SOLVING

Problems are embedded in the fabric of our lives almost to the same extent that decisions are. Every domain of decision making is also a domain in which we have to solve problems. Every decision has an impact on our problems either to minimize or to contribute to them.

Important Facts about Problem Solving

- Some problems solve themselves
- Most problems must be solved one way or the other or else may get worse

Types of Problems

- Problems we ourselves have created by our decisions and behavior (easier to solve by reversing earlier decisions or modifying behavior)
- Problems created by forces outside of us

Each of these can be divided into two groups:

- Problems we can solve in part or whole
- Problems beyond our control, that cannot be solved

Things to avoid in Problem Solving

1. Pseudo-solutions
 - Solutions that seem to solve the problem but they do not.
 - Solutions that solve the problem at the expense of others
2. Pseudo problems
 - These arise when we seek to satisfy false needs and obtain irrational ends.
 - Pseudo problems are dissolved not solved

Big Problems

- Problems for which our responses will have long term consequences
- Problems whose long term consequences must be discovered

Problem Solving Skills

- 1) Regularly rearticulate and re-evaluate our goals, purposes and needs. This is because problems mainly arise because of:
 - Obstacles to reaching our goals or satisfying our needs
 - Misconceptions about our goals purposes
- 2) Identify problems explicitly and analysis them. This enables to avoid being vague about the problem or hiding the problem
- 3) Figure out the information needed and actively seek that information. Almost all the problem solving requires acquisition of key relevant information.
- 4) Carefully analyze, interpret and evaluate the information. This enables one to:
 - To make sense of and give meaning to the information
 - To avoid falsehoods deceptions and distortions
 - To check reliability and relevance of the source
 - To make reasonable inferences
- 5 Figure out the alternatives for action and evaluate them. This is to enable you:
 - To see clearly the different options and inferences the information may lead you to.
 - To make sure that you are not misinterpreting the situation
 - To distinguish what is under your control and what is not
 - To determine both the short term and the long term implications, the limitations I may have etc.
- 6 Taking a strategic approach to the problem
 - Different problems require different strategies e.g. for some it is just wait and see
 - Most cases a more direct strategy is required
- 7 Monitor the implications of your actions as they emerge. This enables one to:
 - Revise strategy or strategies at short moment should the situation so require.
 - Re-look at our strategy and analysis of the problem as more information becomes available
 - Back track, even shift direction if need arises.

COMMON BARRIERS TO PROBLEM SOLVING

Internal Factors

- Ignoring the problem
- Thinking that the problem will go away

- Procrastination
- Thinking that there is no solution
- Fear of failure
- Unwillingness to consider other people's opinions
- Lack of courage to implement the best solution
- Thinking that the solution will take too long, take too much effort

External factors

- Resource constraints
- Resistance to change, new ideas
- Abstract changes, hence need to change strategy

MODELS OF CRITICAL THINKING

SOCRATIC MODEL

Socrates is generally considered to be the father of philosophy. He revolutionized philosophy and gave it its method

Socrates believed that his role in society was to make people examine themselves, because he argued that "an unexamined life was not worthy living".

So he kept on nagging his compatriots, his fellow Athenians to know themselves.

In this regard, Socrates played a twofold role to help others examine themselves:

Role of a Gadfly

-Like a gadfly he kept on stinging and nagging fellow Athenians, not to stop thinking for themselves.

-He argued that an individual who was aware of his ignorance was much better intellectually than who was not because he would strive to know.

Role of Midwife

-Socrates likened his position in society to that of midwife.

-He saw himself as merely assisting others to deliver dormant knowledge or ideas they already had in their minds, in the same way as midwives help mothers to deliver their babies.

-He believed that all people had the capacity to know for themselves what they required was the proper environment and method to realize their potential

Socratic Method/dialectics

-Towards this end he developed a method that came to be known as the Socratic- method, which was essentially interlocutory.

-The Socratic Method is also called Socratic dialect-a method of thinking, which involves the following characteristics

- Dialectical thinking-which involves three main steps: thesis, antithesis and synthesis
- Live exchange and debate of issues- Socrates believed this was the best way to generate knowledge, this way he revolutionized thinking
- Questioning to discover the underlying foundation of things
- Argumentation -to ensure consistence of thought, and avoid contradictions
- Inquiring-to explore all possibilities in thinking

RELEVANCE OF THE SOCRATIC MODEL

- 1) It helps to engage the individual to think for oneself. Socrates believed every person has the capacity to discover knowledge and the truth for oneself
- 2) It takes us through the journey of self-knowledge in order to acquaint with oneself as a conscious, rational being.
- 3) Self-knowledge and exploration in turn reveals self-identity or who am I, what kind of person that we are and what we aspire to be
- 4) It provokes us to questions ourselves, leading to self-discovery of ourselves
- 5) This inquiry into oneself helps to reveal and enhance a person's individuality or the uniqueness of every person.
- 6) It helps to bring out one's potentials and possibilities but also limitations, which is the essence of personal identity, creativity and innovativeness
- 7) Demonstrates the importance of critical thinking in the life of the individual and society.

ON APPEARANCE AND REALITY

Bertrand Russell

-The problem of appearance and reality raises epistemological concerns about our ways of knowing or sources of knowledge.

- When philosophers reflect on the nature of knowledge and the world, they discover that the world is much more sophisticated, complex and even baffling than most people realize.

-The major epistemological question raised is; how can we know anything at all? In the history of Western philosophy this question divided philosophers into two main positions; empiricism and rationalism

Empiricism maintains that we know what we have found out from our senses, and through experience; that is our senses are our main sources of knowledge, thus we know **a posteriori**

Thus rationalism is basically the position that we know what we have thought out; that the mind has the ability to discover knowledge by itself by comparing ideas, that is we know **apriori**

The problem of appearance and is also called the problem of perception and it largely revolves around the (un)reliability of perception (sense derived knowledge) and knowledge gained spontaneously.

Problem of Spontaneity

- Knowledge gained without much reflection often intuitively, or based on first looks (appearances)
- The problem with this is that 'first looks' are not always correct and things are not always what they appear to be.

The Problem of Perception

-Plato captured this problem when he said that "the verdict of senses must be viewed with suspicion and skepticism". And indeed a quick run through the senses reveals vividly the problem of perception.

Sense of Sight

- Tends to depend on a number of factors each of which tends to give a different picture of what we see. For example, the color of objects depends on whether we look at the object with the naked eye or colored glasses
- May also depend on the intensity of the light
- Or whether we have normal sight or color blind
- Or whether we see the object through the microscope or by the naked eye.

Sense of Touch

-The sense of touch tends to depend on how hard we press the object or alternatively, what part of the body presses the object.

-A philosopher, George Berkley once remarked that water may feel warm to one hand and cool to the other if when you plunge both hands into the same water one hand is warm the other is cool.

Sense of Sound

-The sense of hearing presents more or less the same kind of problems. For example if one is standing by a railway line and the whistling locomotive rushes past, the pitch of the whistling tends to change depending on the proximity of the locomotive.

Senses of Smell and Taste

-These too are fraught with perceptual difficulties. They tend to be influenced by what we have just been eating or smelling. For example, cold may influence the taste of food; tea taken immediately after eating a pineapple may not taste the same.

APPEARANCE therefore largely constitutes knowledge or view of the world that we have gained spontaneously without much reflection, through commonsense and sense perception. The problem with this is that appearances can be:

1. **Relative**-vary from one person to the other, from time to time to the same person
2. **Unreliable** as our source of knowledge and judgment
3. **Deceptive**-can appear to be what they are not

-A critical thinker therefore ought to distinguish between appearance and reality to avoid deceptions.

The importance of distinguish between Appearance Reality

-To detect deceptions. We fall victim to deceptions for many reasons. Machiavelli (1469-1527) said “men are so simple in mind and so much dominated by their immediate needs that a deceptive man will always find plenty who are willing to be deceived.

- To have a better understanding of world around us. The world is much more complex than it appears, hence the need to distinguish the two.
- To make correct judgments – we often make wrong decision and judgments due to our inability to distinguish between appearance and reality e.g. advertisements, politics etc.
- To avoid being victims of conmen and women. The basic tool of a con person is to present appearance as if it were either truth or reality
- To detect dogmatic beliefs which greatly hinder our understanding of the world, such beliefs stop us from being inquisitive and rational – depend on blind faith e.g. doomsday cults
- To identify slogans and propaganda and treat them for what they are; slogans are catch phrases that are meant to mobilize people emotionally. They can be misleading, deceptive, detract people from real issues – hence lead people astray. They may make us make wrong decisions and arrive at incorrect conclusions.
- The capacity to distinguish between appearance and reality enhances our power to be open minded and see things more objectively

INDIVIDUALITY AND SOCIALITY

John S. Mill

- John S Mill – English philosopher a champion of individual liberty, freedom or autonomy. The central idea in Mill’s essay is respect for human beings and their fundamental liberties.
- Mill argues that freedom is something good, not because it contributed to an ulterior end but because freedom is the proper condition of a responsible human being.
- Mill believed that to live one’s own life, developing one’s own natural traits and capacities is not just a means to happiness but was a substantive part of happiness.
- A good society must therefore permit freedom to flourish by opening up opportunities for free and satisfying ways of life.
- Liberty is not only an individual good; it is also a social good. Therefore, society that undermines individual freedom not only undermines the good of the individual but its wellbeing.
- According to Mill, “all mankind has no right to silence one dissenter.” This affirms his belief in the freedom of judgment, the right to be convinced rather than coerced
- A good society should therefore not merely tolerate individual freedom; it must work at promoting these freedoms for its own betterment.
- This essay therefore focuses on the relationship between individuality and sociality.

What is individuality?

- The other word for individuality is egoity
- It means the uniqueness of each individual in the way they view the world in a peculiar way as a kind
- Refers to unique personality of each individual – no one is a duplicate of any other person
- Individuality is state of being characterized by independence and autonomy in thought and action.
- Individuality can generally be defined as the irreplaceable characteristics which makes each person unique. It therefore refers to the ability to exercise independence of thought and action.

- It is approaching one's life as an individual's project, to be planned and carried out through; it requires a corresponding set of personal qualities rather than communally minded way of life. It requires critical thinking
- Individuals need to make their own decisions about what they do and how they live rather than allowing their behavior to be dictated.
- Mill believes that people should be left to think, speak as they like; choose their plan of life /way of life and be able to choose their own associates.

What is Sociality?

- This means coexistence with others in a social context. No individual is self-sufficient, so needs others to realize oneself. Hence man is defined variously as rational, moral and as social being.
- **Freedom** is therefore central to individuality and one of the greatest ingredients of human wellbeing. To Mill there are three main domains of human liberty; freedom.

1 Inward domain of consciousness

- Liberty of conscience, thought, feeling
- Liberty of opinion – expressing and publishing opinions
- Thus all opinions are equal, regardless of their origin, until otherwise proved
- Matters of truth or what is right is not dependent on numbers

2 Liberty of Tastes and their Pursuit

- Of framing the plans of our lives to suit our own character and personality
- This involves pursuit of opinion in action.

3 Liberty of combining with other individuals

- Freedom to unite, freedom of association – as long as in uniting people are not deceived or coerced
- Mill believes that people should be left to think, speak as they like; choose their plan of life /way of life and be able to choose their own associates. There are however, many obstacles to individuality, for example:
 - **Authoritarianism** – society or its agents imposing its will on individuals demanding total conformity.

- **Paternalism** – Society acting like father to his children giving them what they need without giving them an opportunity to make their own choices and take personal responsibility.
- **Indoctrination** – placing the determination of one's life in another person by shaping his/her thinking in some pre– conceived manner.

Importance of Individuality to Individual wellbeing

- Freedom – is central to individuality and one of the greatest ingredients of human wellbeing.
- Individuality – involves refinement of the individual's unique character, qualities and contributes to human wellbeing.
- Thus through cultivation of individuality we become better human persons, more refined.
- Individuality develops the best of our qualities and hence enables us to realize our potentials, because we can:
 - plan our own lives
 - attain self- realization
 - take charge of ourselves
- It is a source of happiness and successful living and fulfillment

Importance of Individuality to Societal wellbeing

- Individuality is synonymous with development or leads to societal development
- When there is fullness of life in individuals there is a corresponding degree of the same in society.
- Development of personal qualities in turn makes individuals appreciate others and become more valuable to others
- Individuality cultivates criticality, creativity and innovation, hence may help to;

- Prescribe new ways of doing things
- Set new precedents e.g. in values
- Make discoveries that may uplift society
- May discover new truths and correct society

Therefore, society that silences critical voices therefore undermines its own good as moderate or average men are unlikely to do anything unusual.

CARTESIAN MODEL: THINKING FOR ONESELF

-The discussion is based on Meditations¹ and 2 by Rene Descartes a French philosopher, also well-known mathematician and physicist.

-Descartes meditations are concerned with the foundations of knowledge, thus raising an epistemological question, reliability of our knowledge.

-How secure are the foundations of our knowledge of our world and ourselves?

- Is there any knowledge that is free from doubt, that is indubitable?
- How reliable are the various sources of our knowledge of our world and ourselves?
- He was looking for one truth upon which he would develop all knowledge, the foundation of all knowledge.

Cartesian Methodic Doubt

In the search the foundation of knowledge Descartes developed a method which came to be known as the Cartesian method or Methodic doubt.

The methodic doubt comprises 2 components; inquiry and doubt

Inquiry (which in turn comprises four elements)

1. Not taking anything for granted, accepting only that which is well reasoned, justified, grounded, clear and distinct to the mind
2. To analyze the problem, see the issue or problem in its constituent parts, parts may have characteristics not present the whole. To handle problem systematically.
3. To synthesize-synthesize involves the attempt to find or construct a solution, beginning with the simplest to the most complex, seeing the problem in its totality and how parts relate to the whole.
4. Revision-identify flaws and errors of omission and commission.

Doubt

- Suspending all received opinion that one has the slightest reason to doubt.
- Suspending of all decisions or judgement until we are sure, it is clear and distinct to mind.
- Doubting everything received from the senses until we have strong reason to belief.

How Descartes Demonstrates Thinking for Oneself.

Meditations 1

- He begins by questioning all that his teachers taught him, suspending all received opinion.
- He also doubted what his senses made him to belief as true.
- He further questioned the existence of all objects around him including figures, quantity, shapes, even his own body.

This was after realization that:

- One could have built knowledge on false or unverified beliefs, claims
- All the above were also subject of dreams-dreaming experienced.

Examples

- Illusions-poor man dreaming that he has become a rich man or king
- Captive enjoying illusory liberty-fears to wake up to reality.
- Wax-that changes form in different conditions

Meditations II

In the end, Descartes applies his methodic doubt to arrive at what he thought was indubitable truth - mathematical truth. He thought such truth is found in mathematics.

- Mathematical truth seems indubitable.
- Appeared to be clear and distinct to the mind.
- Mathematics is analytical-talks about simple and general things.
- Not interested in establishing whether they exist or not.
- He thought that whether awake or not asleep- $2+2=4$ and Square has 4 equal sides

But he remembered that even mathematical truth is not free from doubt, why?

- An all-powerful god may have made us such that every time we say $2+2=4$ we err or we deceive ourselves.
- However, Descartes remembered that God is also good, and cannot possibly deceive
- But- there is possibility that we are victims of an evil genius, who has deceived us such that there is no external world, and $2+2=4$ is an error.

Descartes conclusion is that the only indubitable truth is that that i think therefore iam,
Cogito ergo sum

IMPORTANCE OF DESCARTES' MODEL

1. He demonstrates that to steer thought and reason to new directions involves rejection of revolutionizing thinking in which one must stand out as an individual, autonomous thinker.
2. He demonstrates that we must not be captives to traditions, (he was rebelling against old dogmatism of the church). There are many parallels to this dogmatism today such as tribe, political party, gender, religion and so on.
3. He further demonstrates that man through his own faculties can acquire knowledge. This underscores man's rational nature. This however requires clarity of mind and needs a method-it cannot be done haphazardly.
4. We need to be critical, creative, innovative to have to have a clear understanding of the world-demonstrates the importance of critical and creative thinking

PRACTICALITY OF THINKING

Kwasi Wiredu

- This discussion is based on the e s s a y , what can philosophy do for Africa; Kwasi Wiredu wants to demonstrate the practicality of philosophy and critical thinking.
- Philosophy and critical thinking have often been undermined – ostensibly because they are not practical. This practicality is largely understood in terms of utility often measured in narrow, material or economic terms
- In this essay, Wiredu wants to show that this conception view i.e. defining practicality in narrow material economic sense is erroneous based on a misconception of the meaning and task philosophy and the real meaning of practicality.
- Bertrand Russell – argues that the charge that philosophy is not practical is informed by a misconception of the definition of practicality
- He argues that practicality is (and development) – not just about the acquisition of material things. It also means securing of such conditions that permit the self-realization of man as a rational being (mind body). In other words, we must address both the mind and the bodily.
- Jesus is quoted to have wondered; what would benefit man if he were to gain the whole world but lose his soul?

- Form the philosophical point of view one stands to lose his/her soul by:
 - Not thinking for oneself.
 - Being rendered unable to think for oneself.
- Critical thinking enables us to take charge of our own thinking and also resist being rendered unable to think for ourselves
- So the question; what can philosophy do for Africa is a microcosm of the larger question, what can philosophy do for society?

What can philosophy do for society?

- Philosophy has over the ages – endeavored to come to terms with the basic problems and issues that confront man and the world in which he lives
- Philosophy raises fundamental questions about reality and attempts to provide answers to those questions.
- Philosophy is concerned with the search for knowledge and understanding the world in a very complex questions that philosopher have endeavored to grapple with from time immemorial
- As Karl Marx reckons the role of philosophy is not just to understand the world, but to change it for better, to revolutionize it for better Philosophy is practical because it contributes revolutionizing our thinking.

What can philosophy do for Africa?

- In this essay - Kwasi Wiredu debunks the view that philosophy is not practical. He demonstrates that the critical function of philosophy has big role to play in Africa and indeed in any society.
- Wiredu observes that Africa is going through revolutionary times, seeking new social-cultural, economic, political modes that are best suited for Africa's development. He argues that to bring about this realization, we need the critical function of philosophy to help unveil, elucidate the fundamental principles that underlie these modes.

1. Political Revolution

1. We need the critical function of philosophy to **reflect on the best political** system, suited for the African dynamics
2. We need the critical function of philosophy in **formulation of ideology** in the positive or generate sense which Wiredu defines as set of ideas about what a good society should be, it gives direction and vision of the destiny of man and society.

How do we promote ideology in positive sense?

- By giving intellectuals their rightful place in formulation of ideology.
 - In Africa it is left to politicians who are ill equipped – intellectually to handle it.
 - They often lack the conceptual and analytical skills to handle this important exercise.
 - Critical Thinking is required if ideologies we formulate can have any real impact on the basic thought habits.
- 3 We need the critical function of philosophy to oppose the rise of ideology in the negative or degenerate sense. Wiredu defines this as a set of dogmas to be imposed in people by the government even by force. In Africa this has been largely associated with military dictatorship and one party dictatorship.

Characterized by following

- stifles innovation/creativity
- promote intolerance
- stifles freedom of association and expression
- strangled and muzzled the voice of intellectuals
- illogicality of intimidating people unto accepting certain positions

Critical and creative thinking – insisting on questioning, encourages honesty of thought and protects people from hypocrisy, slogans, negative ideology.

2. Social/Cultural Revolution

1. Cultural Appraisal

This requires the critical function of philosophy to engage in critical appraisal of our traditional values, norms and cultures – especially in view of emerging challenges, sensibilities and sensitive – such as gender sensitivity rights of minority groups and so on.

2. Problem of identity

Culture plays an important role of defining our identity. It has often been argued that slavery and slave trade and colonialism seriously impacted African identity by undermining African cultures and values.

Slavery and slave trade – took African and deposited them as human cargo throughout the world. Africans confronted hostile and alien western culture that ridiculed African culture.

Colonialism slammed together, western and African cultures. To justify colonialism European missionaries, anthropologists, sociologists, ethnologists created racist mythology – which contrasted whiteness and blackness

- **Whiteness represents** humanity, modernity, rationality, science
- **Blackness represents** barbarism, slavery, primitiveness, mysticism, superstition, irrationality and so on

These happenings brutally and systematically attempted to annihilate indigenous African cultures – by imposing European cultures, making Europe the yardstick against which to measure human worth.

These experiences implanted a psychological inferiority complex which made African ashamed of themselves, of their blackness

Critical and creative thinking, the critical function of philosophy can help us to

- question and refind ourselves
- re-discover ourselves
- redefine ourselves

In conclusion, philosophy has three key characteristics that can be applied in social and cultural appraisal. Philosophy:

- relies on reason
- context – sensitive
- self-corrective

3. Economic Revolution

Africa should embrace and benefit from the economic and technological developments of the West and other parts of the world. Accordingly Africa has been embracing new economic modes and technologies largely borrowed from the West. The critical function of philosophy can help us on evaluating such modes and concepts such as:

- globalization
- free market economies
- economic subsidies
- economic blocks
- reliance on donors

Some of these developments have generally impacted on the poor. Critical thinking can help to unveil and evaluate the fundamental principles and values that underlie these new economic modes and technologies.

Critical function of philosophy will help us to reflect on why Africa has lagged behind in economic and technological development as compared to other countries that were at the same level at independence, despite the fact that Africa has a lot of natural resources.

Critical function of philosophy can help us to reflect on and evaluate the impact of climate change on economies and general ecological wellbeing.